

Quiz 2, ASM

Time allowed: 30 mins

Suppose $X_1, \dots, X_n$ are i.i.d. from a distribution with pdf f . Define

$$\hat{f}(t) = \frac{1}{n} \sum_{i=1}^n K_h(X_i - t),$$

where

$$K_h(x) = \frac{1}{h} K\left(\frac{x}{h}\right), \quad K \geq 0, \quad \int K(x) dx = 1, \quad \int x K(x) dx = 0, \quad \int x^2 K(x) dx = 1.$$

Prove the following:

1.

$$\mathbb{E}[\hat{f}(t)] = f(t) + \frac{h^2}{2} f''(t) + o(h^2).$$

2.

$$\text{Var}(\hat{f}(t)) = \frac{1}{nh^2} \mathbb{E} \left[K^2 \left(\frac{X_1 - t}{h} \right) \right] - \frac{1}{n} f^2(t) + o\left(\frac{h^2}{n}\right).$$

3.

$$\mathbb{E} \left[K^2 \left(\frac{X_1 - t}{h} \right) \right] = h f(t) \int K^2(y) dy + o(h).$$

4.

$$\text{Var}(\hat{f}(t)) = \frac{1}{nh} f(t) \int K^2(y) dy + o\left(\frac{1}{nh}\right).$$

For proving any part, you may use the results of the previous parts even if you could not prove them.